BAYESIAN BIOSTATISTICS / I-SPY2

Modeling breast cancer treatment response

Baseline MRI features, receptor status and uncertainty-aware statistical modeling.

Jennifer Zhu | Original course analysis: March 2026

384	3	8,000
Complete-case observations	Primary model specifications	Post-warmup draws per fitted model

The research question

Do pretreatment tumor longest diameter and sphericity help characterize pathologic complete response (pCR)? Does receptor status provide additional information? This course project examines those questions with interpretable Bayesian logistic regression.

Data and study design

Clinical and MRI-feature tables from the I-SPY2 imaging cohort were joined by participant identifiers. An independent Python audit of the supplied 985-row clinical and 384-row MRI tables confirmed 384 complete cases for pCR, diameter, sphericity, HR and HER2, matching the course report. Both continuous predictors were standardized before modeling.

Modeling strategy

Specification	Predictors
Imaging only	Standardized diameter and sphericity
Additive	Imaging features plus HR and HER2
Interaction	Imaging features plus HR, HER2 and HR x HER2

All models used a Bernoulli likelihood with logit link. Normal priors were specified with SD 1.5 for the intercept, 0.7 for imaging coefficients and 1 for receptor/interaction coefficients. Each fit used four chains, 4,000 iterations per chain, 2,000 warmup and seed 123.

Provenance: model findings come from the archived course PDF. Input counts were verified separately in Python; the reconstructed R/Stan models have not been rerun.

Receptor status carried clearer associations

Conditional on the included predictors, HR-positive status was associated with lower odds of pCR and HER2-positive status with higher odds. The two MRI-feature intervals overlapped an odds ratio of 1.

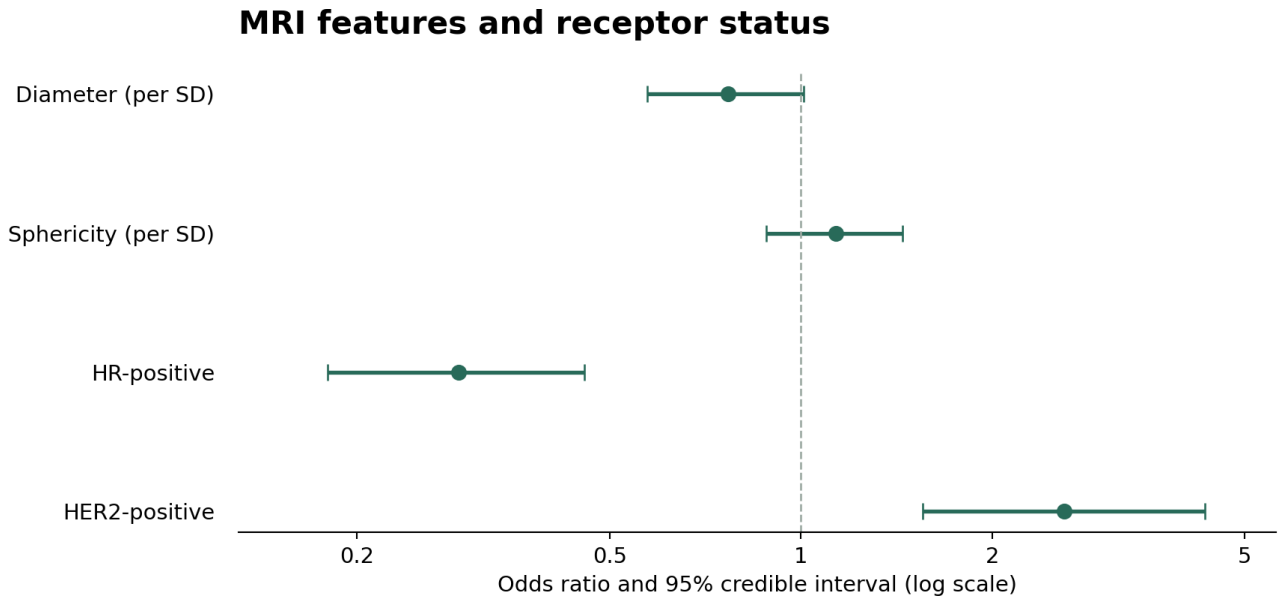


Figure 1. Redrawn from archived additive-model estimates. Points show exponentiated posterior mean log-odds coefficients; bars show 95% credible intervals. No new model was fitted.

Predictor	Odds ratio	95% credible interval
Diameter (per SD)	0.77	0.57 to 1.01
Sphericity (per SD)	1.13	0.88 to 1.45
HR-positive	0.29	0.18 to 0.46
HER2-positive	2.60	1.56 to 4.32

What the estimates mean

The receptor variables provided stronger evidence of association in the additive model than these two baseline size and shape features. This does not establish that MRI is uninformative generally; the analysis included only a small subset of possible imaging measurements.

These conditional associations are not causal treatment effects. OR point estimates follow the source's exp(mean log-odds) convention, rather than the mean of exponentiated posterior draws.

A parsimonious additive specification

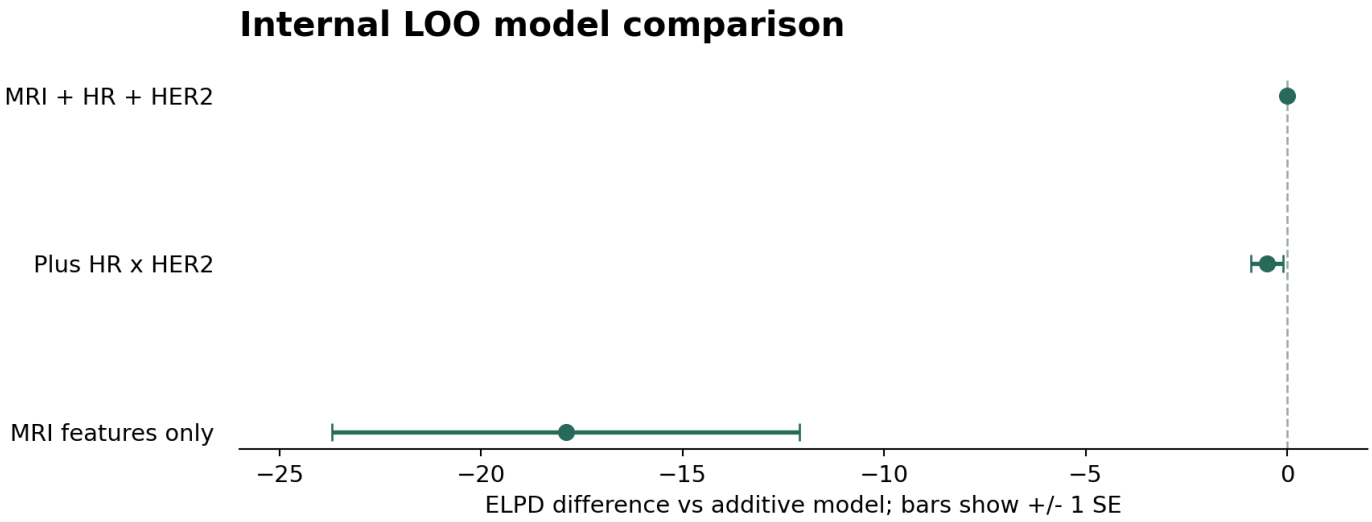


Figure 2. Archived LOO differences relative to the additive model. Error bars are +/- one standard error of the difference, not 95% intervals. The reference model's zero is defined by the comparison.

The additive model ranked highest. The interaction specification had a similar score (difference -0.5, SE 0.4), while imaging alone scored lower (-17.9, SE 5.8). The original analysis selected the additive model for its balance of fit and simplicity.

Diagnostics and sensitivity

The archived additive-model summary reported R-hat values of 1.00 and large effective sample sizes. A default-prior refit retained similar coefficient directions: HR OR 0.26 and HER2 OR 2.86, compared with 0.29 and 2.60 under the specified priors. This is a limited sensitivity check.

Limits of the evidence

Complete-case selection, a small predictor set, and lack of treatment-arm adjustment limit interpretation. The comparison used internal LOO, not external validation, and preserved global standardization. Original draws and full Pareto-k output were unavailable. The source PDF also retains compiler messages alongside later sampling outputs; an error-free original execution is not assumed.

Reconstruction and sources

The editable original was unavailable. R scripts were reconstructed from the author's archived report; displayed results were transcribed, not refitted. Source: reports/source-course-report.pdf, pp. 3-4, 7-9 and 14-15. Data collection: cancerimagingarchive.net/collection/ispy2/. Software: paulbuerkner.com/brms/.